



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

09-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE (ALL STAR BATCH)]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	A	A	B	D	B	B	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	B	C	C	B	C	B	B	A	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	25	1	3	3	500	4	8	6	200	16

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	B	C	A	C	B	D	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	B	C	D	B	B	D	D	A	D
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	2	2	5	2	2	64	2	173	26	3

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	C	B	D	B	D	A	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	C	B	A	A	C	D	B	D	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	781	7	5	6	26	77	924	1494	2	1250

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (B)

As electric field is in y-direction so electric flux is only due to top and bottom surface

Bottom surface $y = 0$

$$\Rightarrow E = 0 \Rightarrow \phi = 0$$

Top surface $y = 0.5$ m

$$\Rightarrow E = 150 (.5)^2 = \frac{150}{4}$$

$$\text{Now flux } \phi = EA = \frac{150}{4} (.5)^2 = \frac{150}{16}$$

$$\text{By Gauss's law } \phi = \frac{Q_{\text{in}}}{\epsilon_0}$$

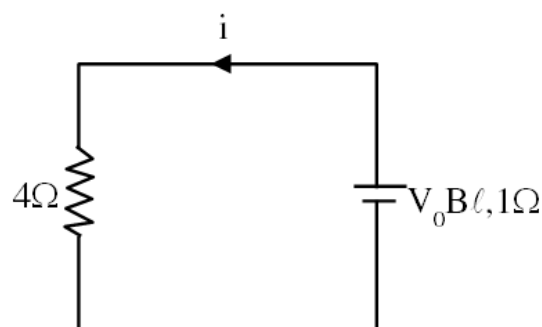
$$\frac{150}{16} = \frac{Q_{\text{in}}}{\epsilon_0}$$

$$Q_{\text{in}} = \frac{150}{16} \times 8.85 \times 10^{-12} = 8.3 \times 10^{-11} \text{ C}$$

Option (2)

2. Ans (B)

Equivalent circuit



$$i = \frac{V_0 B l}{4 + 1} \Rightarrow V_0 = \frac{5 (2 \text{ mA})}{5 \times .2} = 10^{-2} \text{ m/s} = 1 \text{ cm/s}$$

Option (2)

3. **Ans (A)**

De-Broglie wavelength

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mE}}$$

Where E is kinetic energy

$$E = \frac{3kT}{2} \text{ for gas}$$

$$\lambda = \frac{h}{\sqrt{3mkT}} = \frac{6.6 \times 10^{-34}}{\sqrt{3 \times 9 \times 10^{-31} \times 1.38 \times 10^{-23} \times 300}}$$

$$\lambda = 6.26 \times 10^{-9} \text{ m} = 6.26 \text{ nm}$$

Option (1)

4. **Ans (A)**

Work done = Change in kinetic energy

$$W_{mg} + W_{\text{air-friction}} = \frac{1}{2}m(.8\sqrt{gh})^2 - \frac{1}{2}m(0)^2$$

$$W_{\text{air-friction}} = \frac{.64}{2}mgh - mgh = -0.68mgh$$

Option (1)

5. **Ans (B)**

$$\text{Range } R = \frac{u^2 \sin 2\theta}{g} \text{ and same for } \theta \text{ and } 90 - \theta$$

So same for 42° and 48°

$$\text{Maximum height } H = \frac{u^2 \sin^2 \theta}{2g}$$

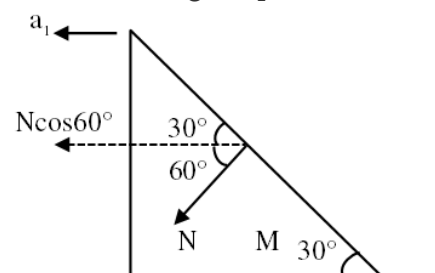
H is high for higher θ

So H for 48° is higher than H for 42°

Option (2)

6. **Ans (D)**

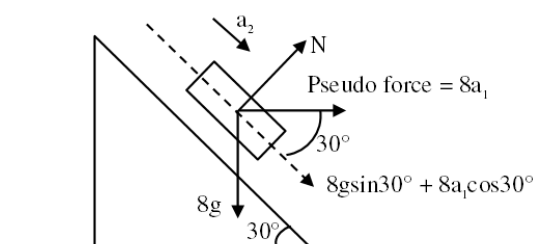
Let acceleration of wedge is a_1 and acceleration of block w.r.t. wedge is a_2



$$N \cos 60^\circ = Ma_1 = 16a_1$$

$$\Rightarrow N = 32a_1$$

F.B.D. of block w.r.t wedge



$\perp$ to incline

$$N = 8g \cos 30^\circ - 8a_1 \sin 30^\circ \Rightarrow 32a_1 = 4\sqrt{3}g - 4a_1$$

$$\Rightarrow a_1 = \frac{\sqrt{3}}{9}g$$

Along incline

$$8g \sin 30^\circ + 8a_1 \cos 30^\circ = ma_2 = 8a_2$$

$$a_2 = g \times \frac{1}{2} + \frac{\sqrt{3}}{9}g \cdot \frac{\sqrt{3}}{2} = \frac{2g}{3}$$

Option (4)

7. **Ans (B)**

$$\text{mass of ice } m = \rho A \ell = 10^3 \times 10^{-4} \times 1 = 10^{-1} \text{ kg}$$

Energy required to melt the ice

$$Q = ms\Delta T + mL$$

$$= 10^{-1} (2 \times 10^3 \times 10 + 3.33 \times 10^5) = 3.53 \times 10^4 \text{ J}$$

$$Q = i^2 RT \Rightarrow 3.53 \times 10^4 = \left(\frac{1}{2}\right)^2 (4 \times 10^3)(t)$$

$$\text{Time} = 35.3 \text{ sec}$$

Option (2)

8. **Ans (B)**

$$y = \frac{MgL^3}{4bd^3\delta}$$

$$\frac{\Delta y}{y} = \frac{\Delta M}{M} + \frac{3\Delta L}{L} + \frac{\Delta b}{b} + \frac{3\Delta d}{d} + \frac{\Delta \delta}{\delta}$$

$$\frac{\Delta y}{y} = \frac{10^{-3}}{2} + \frac{3 \times 10^{-3}}{1} + \frac{10^{-2}}{4} + \frac{3 \times 10^{-2}}{4} + \frac{10^{-2}}{5}$$

$$= 10^{-3} [0.5 + 3 + 2.5 + 7.5 + 2] = 0.0155$$

Option (2)

9. **Ans (C)**

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{eq}} = \frac{1}{4} + \frac{1}{4} \Rightarrow R_{eq} = 2\Omega$$

$$\text{Also } \frac{\Delta R_{eq}}{R_{eq}^2} = \frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}$$

$$\frac{\Delta R_{eq}}{4} = \frac{.8}{16} + \frac{.4}{16} = \frac{1.2}{16}$$

$$\Delta R_{eq} = 0.3\Omega$$

$$R_{eq} = (2 \pm 0.3)\Omega$$

Option (3)

10. **Ans (B)**

$$(t_{1/2})_x = (\tau)_y$$

$$\Rightarrow \frac{\ell n 2}{\lambda_x} = \frac{1}{\lambda_y} \Rightarrow \lambda_x = 0.693 \lambda_y$$

$$\text{Also initially } N_x = N_y = N_0$$

$$\text{Activity } A = \lambda N$$

$$\text{As } \lambda_x < \lambda_y \Rightarrow A_x < A_y$$

$\Rightarrow y$ will decay faster than x

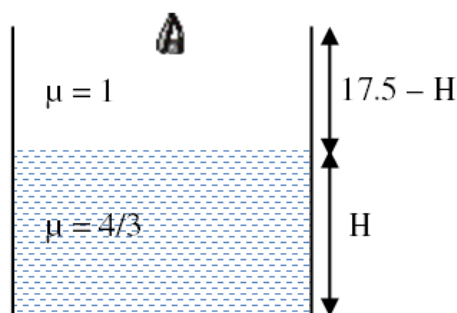
Option (2)

11. **Ans (A)**

Conceptual question

Option (1)

12. **Ans (B)**



Height of water observed by observer

$$= \frac{H}{\mu_w} = \frac{H}{(4/3)} = \frac{3H}{4}$$

Height of air observed by observer = $17.5 - H$

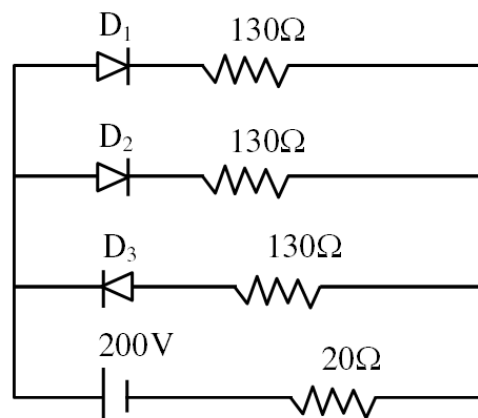
According to question, both height observed by observer is same.

$$\frac{3H}{4} = 17.5 - H$$

$$\Rightarrow H = 10 \text{ cm}$$

Option (2)

13. **Ans (C)**



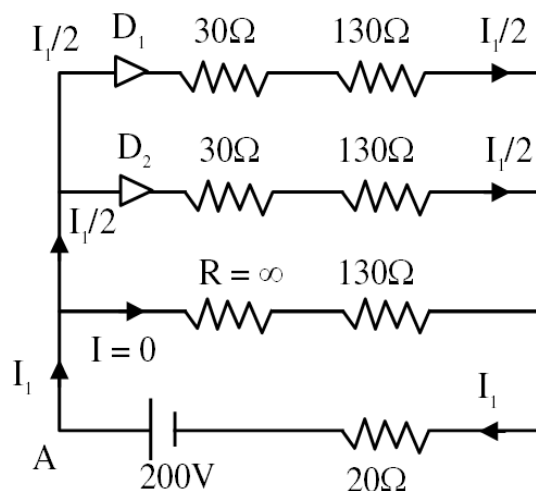
As per diagram,

Diode D_1 & D_2 are in forward bias i.e. $R = 30\Omega$

whereas diode D_3 is in reverse bias i.e. $R = \text{infinite}$

$\Rightarrow$ Equivalent circuit will be

Applying KVL starting from point A



$$- \left(\frac{I_1}{2} \right) \times 30 - \left(\frac{I_1}{2} \right) \times 130 - I_1 \times 20 + 200 = 0$$

$$\Rightarrow -100 I_1 + 200 = 0$$

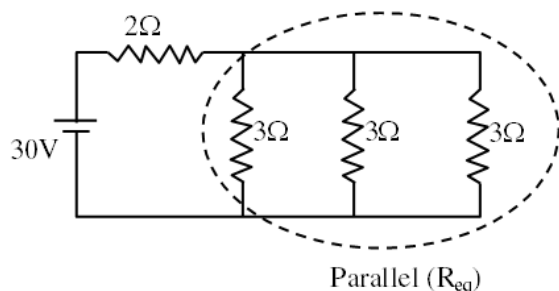
$$I_1 = 2$$

Option (3)

14. **Ans (C)**

In steady state, inductor behaves as a conducting wire.

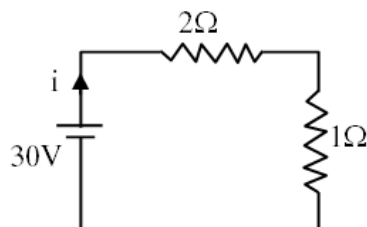
So, equivalent circuit becomes



$$\frac{1}{R_{eq}} = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$

$$\Rightarrow R_{eq} = 1\Omega$$

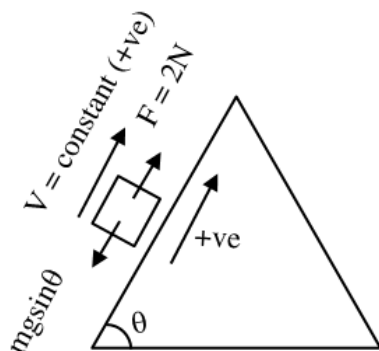
$\Rightarrow$ Circuit becomes



$$\Rightarrow i = \frac{30}{3} = 10A$$

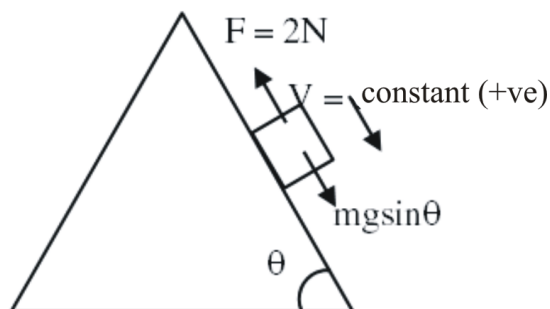
15. **Ans (B)**

During upward motion



$$F = 2N = (+ve) \text{ constant}$$

During downward motion



$$\Rightarrow F = 2N = (-ve) \text{ constant}$$

$\Rightarrow$ Best possible answer is option (2)

16. **Ans (C)**

From potential energy curve

$$U_{\max} = \frac{1}{2}kA^2 \Rightarrow 10 = \frac{1}{2}k(2)^2$$

$$\Rightarrow k = 5$$

$$\text{Now } T_{\text{spring}} = T_{\text{pendulum}}$$

$$2\pi\sqrt{\frac{5}{g}} = 2\pi\sqrt{\frac{4}{g}}$$

$$\Rightarrow 1 = \sqrt{\frac{4}{g}} \Rightarrow g = 4 \text{ on planet}$$

Option (3)

17. **Ans (B)**

$$V = V_0 (1 - e^{-t/RC})$$

$$2 = 20 (1 - e^{-t/RC})$$

$$\frac{1}{10} = 1 - e^{-t/RC}$$

$$e^{-t/RC} = \frac{9}{10}$$

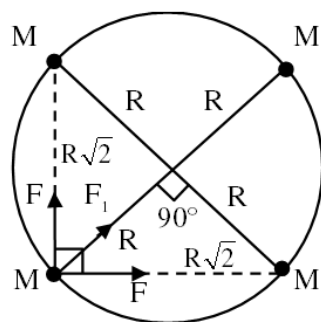
$$e^{t/RC} = \frac{10}{9}$$

$$\frac{t}{RC} = \ln\left(\frac{10}{9}\right) \Rightarrow C = \frac{t}{R \ln\left(\frac{10}{9}\right)}$$

$$C = \frac{10^{-6}}{10 \times .105} = .95\mu F$$

Option (2)

18. Ans (B)



$$F_{\text{net}} = \frac{MV^2}{R}$$

$$\sqrt{2}F + F_1 = \frac{MV^2}{R}$$

$$\sqrt{2} \frac{GMM}{(\sqrt{2}R)^2} + \frac{GMM}{(2R)^2} = \frac{MV^2}{R}$$

$$\frac{GM}{R} \left(\frac{1}{\sqrt{2}} + \frac{1}{4} \right) = V^2$$

$$\frac{GM}{R} \left(\frac{4 + \sqrt{2}}{4\sqrt{2}} \right) = V^2$$

$$V = \sqrt{\frac{GM(4 + \sqrt{2})}{R4\sqrt{2}}}$$

$$V = \frac{1}{2} \sqrt{\frac{GM(2\sqrt{2} + 1)}{R}}$$

Option (2)

19. Ans (A)

$$\text{Speed of wave} = \frac{2 \times 10^{10}}{200} = 10^8 \text{ m/s}$$

$$\text{Refractive index} = \frac{3 \times 10^8}{10^8} = 3$$

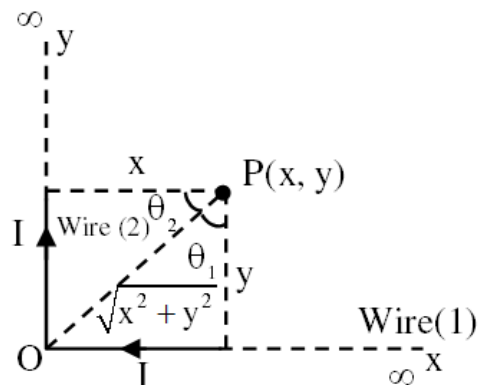
$$\text{Now refractive index} = \sqrt{\epsilon_r \mu_r}$$

$$3 = \sqrt{\epsilon_r} \quad (1)$$

$$\Rightarrow \epsilon_r = 9$$

Option (1)

20. Ans (A)



$$B_{\text{due to wire (1)}} = \frac{\mu_0 I}{4\pi y} [\sin 90^\circ + \sin \theta_1]$$

$$= \frac{\mu_0 I}{4\pi y} \left(1 + \frac{x}{\sqrt{x^2 + y^2}} \right) \dots\dots(1)$$

$$B_{\text{due to wire (2)}} = \frac{\mu_0 I}{4\pi x} (\sin 90^\circ + \sin \theta_2)$$

$$= \frac{\mu_0 I}{4\pi x} \left(1 + \frac{y}{\sqrt{x^2 + y^2}} \right) \dots\dots(2)$$

Total magnetic field

$$B = B_1 + B_2$$

$$B = \frac{\mu_0 I}{4\pi} \left[\frac{1}{y} + \frac{x}{y\sqrt{x^2 + y^2}} + \frac{1}{x} + \frac{y}{x\sqrt{x^2 + y^2}} \right]$$

$$B = \frac{\mu_0 I}{4\pi} \left[\frac{x + y}{xy} + \frac{x^2 + y^2}{xy\sqrt{x^2 + y^2}} \right]$$

$$B = \frac{\mu_0 I}{4\pi} \left[\frac{x + y}{xy} + \frac{\sqrt{x^2 + y^2}}{xy} \right]$$

$$B = \frac{\mu_0 I}{4\pi xy} [\sqrt{x^2 + y^2} + (x + y)]$$

Option (1)

PART-1 : PHYSICS

SECTION-II

1. Ans (25)

Pressure is not changing $\Rightarrow$ isobaric process

$$\Rightarrow \Delta U = nC_v \Delta T = \frac{5nR\Delta T}{2}$$

$$\text{and } W = nR\Delta T$$

$$\frac{\Delta U}{W} = \frac{5}{2} = \frac{x}{10} \Rightarrow x = 25.00$$

2. Ans (1)

Given amplitude $\propto$ slit width

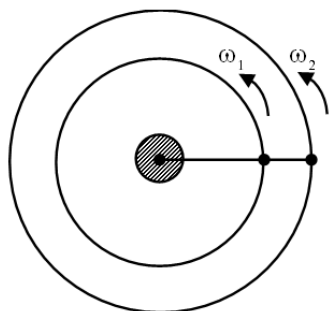
Also intensity $\propto$ (Amplitude) $^2 \propto$ (Slit width) 2

$$\frac{I_1}{I_2} = \left(\frac{3}{1}\right)^2 = 9 \Rightarrow I_1 = 9I_2$$

$$\frac{I_{\min}}{I_{\max}} = \left(\frac{\sqrt{I_1} - \sqrt{I_2}}{\sqrt{I_1} + \sqrt{I_2}}\right)^2 = \left(\frac{3-1}{3+1}\right)^2 = \frac{1}{4} = \frac{x}{4}$$

$$\Rightarrow x = 1.00$$

3. Ans (3)



$$T_1 = 1 \text{ hour}$$

$$\Rightarrow \omega_1 = 2\pi \text{ rad/hour}$$

$$T_2 = 8 \text{ hours}$$

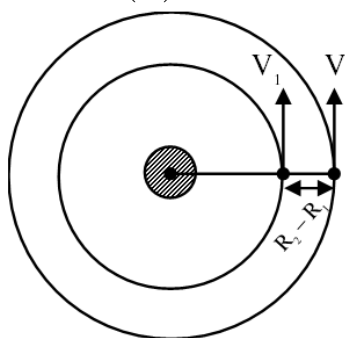
$$\Rightarrow \omega_2 = \frac{\pi}{4} \text{ rad/hour}$$

$$R_1 = 2 \times 10^3 \text{ km}$$

$$\text{As } T^2 \propto R^3$$

$$\Rightarrow \left(\frac{R_2}{R_1}\right)^3 = \left(\frac{T_2}{T_1}\right)^2$$

$$\Rightarrow \frac{R_2}{R_1} = \left(\frac{8}{1}\right)^{2/3} = 4 \Rightarrow R_2 = 8 \times 10^3 \text{ km}$$



$$V_1 = \omega_1 R_1 = 4\pi \times 10^3 \text{ km/h}$$

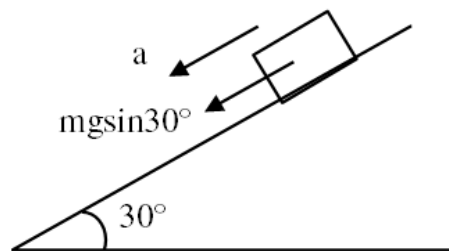
$$V_2 = \omega_2 R_2 = 2\pi \times 10^3 \text{ km/h}$$

$$\text{Relative } \omega = \frac{V_1 - V_2}{R_2 - R_1} = \frac{2\pi \times 10^3}{6 \times 10^3}$$

$$= \frac{\pi}{3} \text{ rad/hour}$$

$$x = 3$$

4. Ans (3)

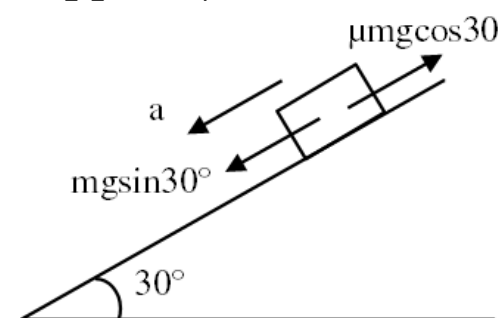


On smooth incline

$$a = g \sin 30^\circ$$

$$\text{by } S = ut + \frac{1}{2}at^2$$

$$S = \frac{1}{2} \frac{g}{2} T^2 = \frac{g}{4} T^2 \dots\dots(i)$$



On rough incline

$$a = g \sin 30^\circ - \mu g \cos 30^\circ$$

$$\text{by } S = ut + \frac{1}{2}at^2$$

$$S = \frac{1}{4} g (1 - \sqrt{3}\mu) (\alpha T)^2 \dots(ii)$$

By (i) and (ii)

$$\frac{1}{4} g T^2 = \frac{1}{4} g (1 - \sqrt{3}\mu) \alpha^2 T^2$$

$$\Rightarrow 1 - \sqrt{3}\mu = \frac{1}{\alpha^2} \Rightarrow \mu = \left(\frac{\alpha^2 - 1}{\alpha^2}\right) \cdot \frac{1}{\sqrt{3}}$$

$$\Rightarrow x = 3.00$$

5. Ans (500)

Given

Translation K.E. of N_2 = K.E. of electron

$$\frac{3}{2} kT = eV$$

$$\frac{3}{2} \times 1.38 \times 10^{-23} T = 1.6 \times 10^{-19} \times 0.1$$

$$\Rightarrow T = 773 \text{ K}$$

$$T = 773 - 273 = 500^\circ \text{C}$$

6. Ans (4)

$$\text{First case } P_1 = \frac{V^2}{R} = \frac{(240)^2}{36}$$

Second case Resistance of each half = 18 Ω

$$P_2 = \frac{(240)^2}{18} + \frac{(240)^2}{18} = \frac{(240)^2}{9}$$

$$\frac{P_1}{P_2} = \frac{1}{4}$$

$$x = 4.00$$

7. Ans (8)

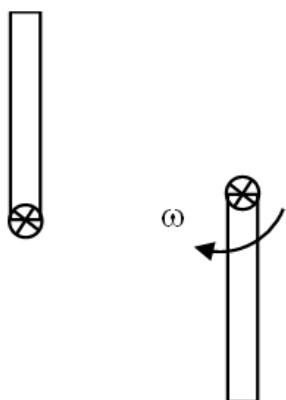
$$\text{Thermal force } F = Ay\alpha\Delta T$$

$$F = (10 \times 10^{-4}) (2 \times 10^{11}) (10^{-5})(400)$$

$$F = 8 \times 10^5 \text{ N}$$

$$\Rightarrow x = 8$$

8. Ans (6)



$$\text{by energy conservation } mgl = \frac{1}{2}I\omega^2 = \frac{1}{2} \frac{ml^2}{3} \omega^2$$

$$\Rightarrow \omega = \sqrt{\frac{6g}{l}}$$

$$\text{Speed } v = \omega r = \omega l = \sqrt{6gl}$$

$$v = \sqrt{6 \times 10 \times 6} = 6 \text{ m/s}$$

9. Ans (200)

$$A_{\max} = A_C + A_m = 250 + 150 = 400$$

$$A_{\min} = A_C - A_m = 250 - 150 = 100$$

$$\frac{A_{\min}}{A_{\max}} = \frac{100}{400} = \frac{1}{4} = \frac{50}{200}$$

$$x = 200$$

10. Ans (16)

$$\text{Work} = \Delta K.E.$$

$$W_{\text{friction}} + W_{\text{spring}} = 0 - \frac{1}{2}mv^2$$

$$-\frac{90}{100} \left(\frac{1}{2}mv^2 \right) + W_{\text{spring}} = -\frac{1}{2}mv^2$$

$$W_{\text{spring}} = -\frac{10}{100} \times \frac{1}{2}mv^2$$

$$-\frac{1}{2}kx^2 = -\frac{1}{20}mv^2$$

$$\Rightarrow k = \frac{40000 \times (20)^2}{10 \times (1)^2} = 16 \times 10^5$$

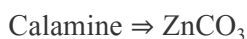
PART-2 : CHEMISTRY

SECTION-I

1. Ans (C)

Clean water could have BOD value of less than 5 ppm whereas highly polluted water could have a BOD value of 17 ppm or more.

2. Ans (B)



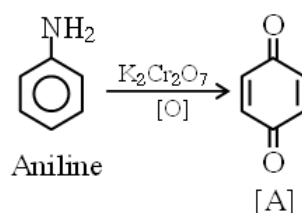
3. Ans (B)

Solution NaH_2 is not reducing agent

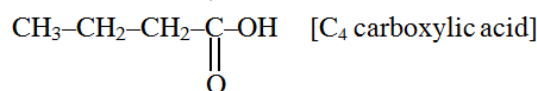
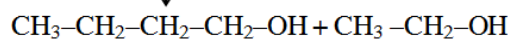
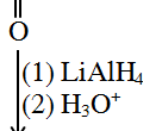
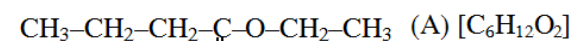
4. Ans (C)

By observation we get this plot during measurable temperatures

5. Ans (A)



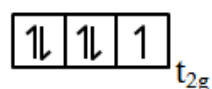
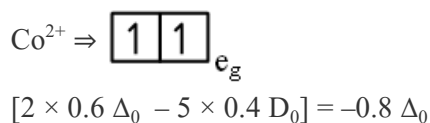
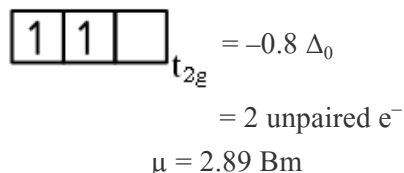
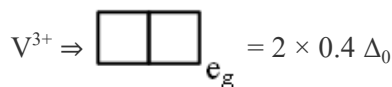
6. Ans (C)



7. **Ans (B)**

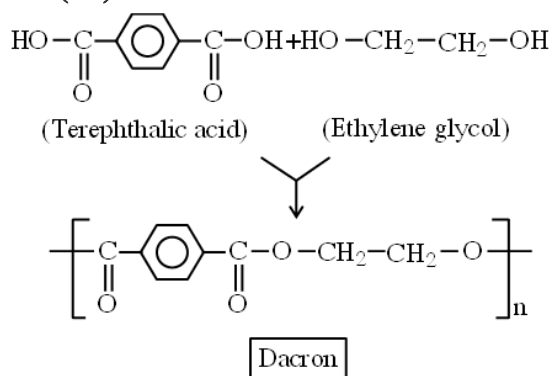
According to type of reactions for preparation, colloids have been classified

8. **Ans (D)**



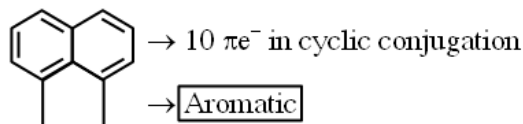
3 unpaired $e^- \Rightarrow \mu = 3.87 \text{ BM}$
hence d^7 configuration is of Co^{2+}

9. **Ans (B)**



10. **Ans (D)**

(1) (Acenaphthene)



(2) $\square$ $\rightarrow 4 \pi e^-$ in ring conjugation
 $\Rightarrow$ Anti Aromatic

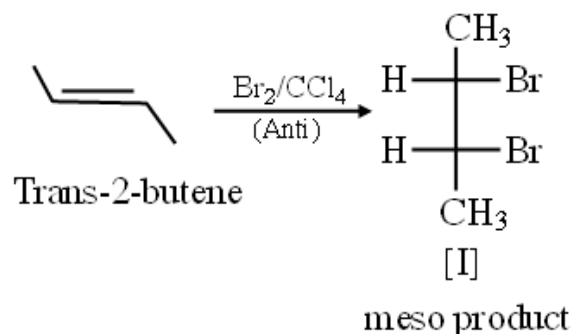
(3) $\text{Cyclopentadienyl}^+ \rightarrow 4 \pi e^-$ in ring conjugation
 $\Rightarrow$ Antiaromatic

(4) $\text{Cyclopentadienyl}^- \Rightarrow 6 \pi e^-$ in ring conjugation $\Rightarrow$ **Aromatic**
Cyclopentadienyl anion

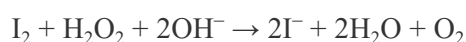
11. **Ans (D)**

Colour of Fe^{2+} is observed green and Fe^{3+} is yellow

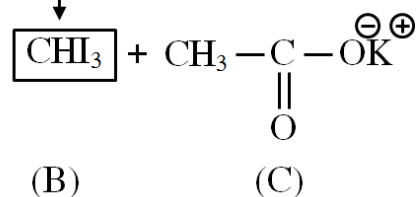
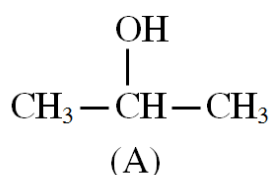
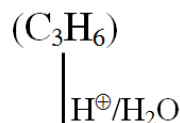
12. **Ans (B)**



13. **Ans (C)**

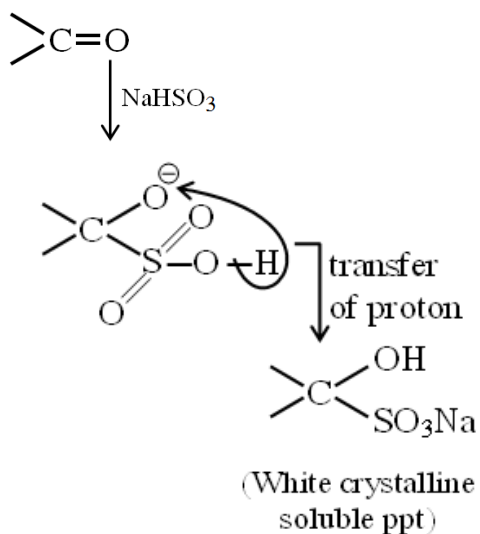


14. **Ans (D)**

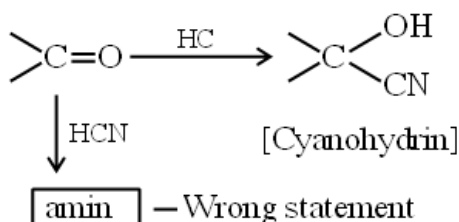


15. Ans (B)

Statement I : Correct



Statement II :

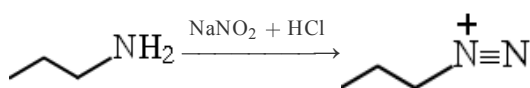


amin – Wrong statement

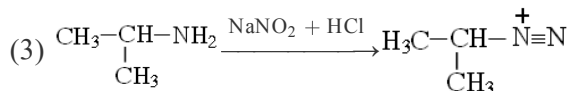
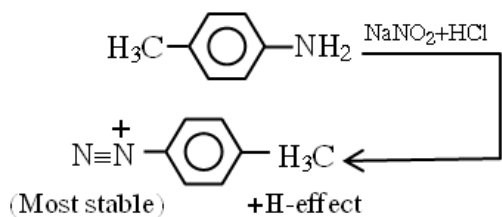
(Amine not formed)

16. Ans (B)

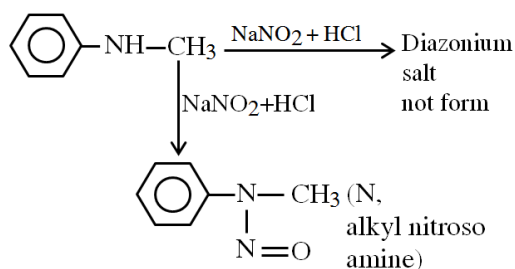
(1)



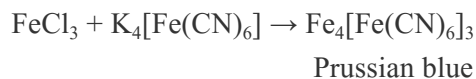
(2)



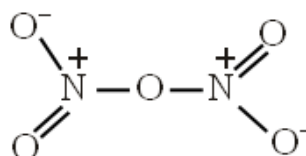
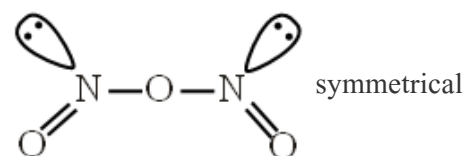
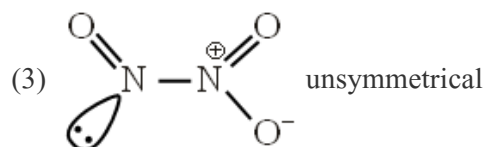
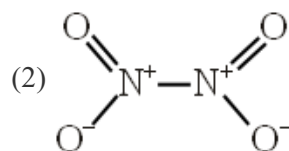
(4)



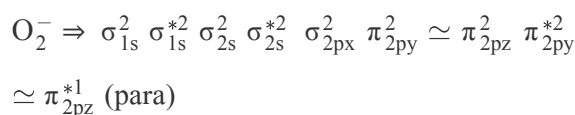
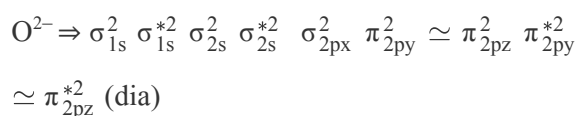
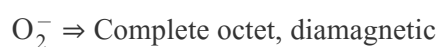
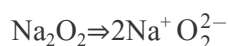
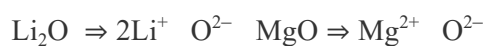
17. Ans (D)



18. Ans (D)



19. Ans (A)



20. Ans (D)



PART-2 : CHEMISTRY

SECTION-II

1. **Ans (2)**

Initial mass of gas = 29 - 14.8 = 14.2 Kg

mass of gas used = 29 - 23 = 6 Kg

gas left = 14.2 - 6 = 8.2 Kg

$$(1) 3.47 \times V = \left(\frac{14.2 \times 10^3}{M} \right) \times R \times T$$

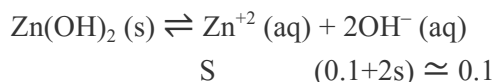
$$(2) p \times V = \left(\frac{8.2 \times 10^3}{M} \right) \times R \times T$$

Divide :

$$\frac{(1)}{(2)} \Rightarrow \frac{3.47}{P} = \frac{14.2}{8.2}$$

$$P = 2.003$$

2. **Ans (2)**



$$K_{sp} = S(0.1)^2$$

$$2 \times 10^{-20} = S \times 10^{-2} \Rightarrow S = 2 \times 10^{-18}$$

$$= x \times 10^{-18}$$

$$x = 2$$

3. **Ans (5)**

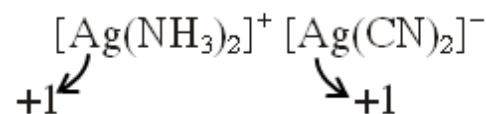
$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G = 57.8 - \frac{298(-176)}{1000}$$

$$\Delta G = -5.352 \text{ kJ/mole}$$

[Nearest integer value] = 5

4. **Ans (2)**



5. **Ans (2)**

$$\text{No. of atoms} = \frac{8}{23} \times 6.02 \times 10^{23}$$

$$= 2.09 \times 10^{23}$$

$$\approx 2 \times 10^{23}$$

$$= x \times 10^{23}$$

$$x = 2$$

6. **Ans (64)**

$$\text{Moles of } \text{CuSO}_4 \cdot 5\text{H}_2\text{O} = \frac{80}{249.54}$$

$$\text{Molarity} = \frac{\frac{80}{249.54}}{5} = 64.117 \times 10^{-3}$$

Nearest integer, x = 64

7. **Ans (2)**

Total energy per sec. = 50 J

$$50 = \frac{n \times 6.63 \times 10^{-34} \times 3 \times 10^8}{795 \times 10^{-9}}$$

n = 1998.49 $\times 10^{17}$ [n = no. of photons per second]

$$= 1.998 \times 10^{20}$$

$$\approx 2 \times 10^{20}$$

$$= x \times 10^{20}$$

$$x = 2$$

8. **Ans (173)**

$$\text{B}_2^+ \Rightarrow \sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2py}^1 \approx \pi_{2pz}^0$$

$$\Rightarrow 9e^-$$

$$\mu = \sqrt{1(1+2)} = \sqrt{3} \text{ BM}$$

$$= 1.73 \text{ BM}$$

$$= 1.73 \times 10^{-2} \text{ BM}$$

9. **Ans (26)**

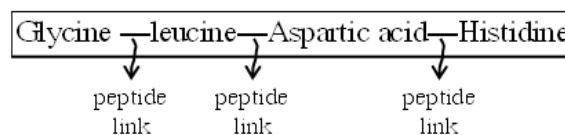
$$k = 1.07 \times 10^6 \text{ Sm}^{-1}, \quad R = 0.243 \text{ W}$$

$$G = \frac{1}{R} = \frac{1}{0.243} \Omega^{-1}$$

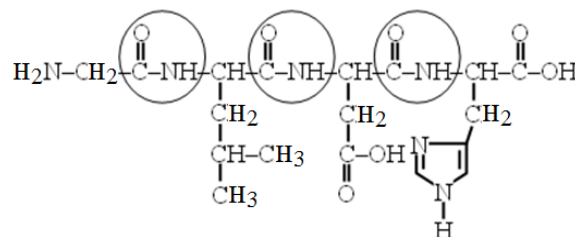
$$k = G \times G^*$$

$$G^* = \frac{k}{G} = \frac{1.07 \times 10^6}{\frac{1}{0.243}} \approx 26 \times 10^4 \text{ m}^{-1}$$

10. **Ans (3)**



Total (3) peptide linkages are present



3 peptide linkage

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (B)**

$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\frac{\pi}{4} \int_2^{\sec^2 x} f(x) dx}{x^2 - \frac{\pi^2}{16}}$$

$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\pi}{4} \cdot \frac{[f(\sec^2 x) \cdot 2 \sec x \cdot \sec x \tan x]}{2x}$$

$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{\pi}{4} f(\sec^2 x) \cdot \sec^3 x \cdot \frac{\sin x}{x}$$

$$\frac{\pi}{4} f(2) \cdot (\sqrt{2})^3 \cdot \frac{1}{\sqrt{2}} \times \frac{4}{\pi}$$

$$\Rightarrow 2f(2)$$

2. **Ans (C)**

$$\cos^{-1}(\cos(-5)) + \sin^{-1}(\sin(6)) - \tan^{-1}(\tan(12))$$

$$\Rightarrow (2\pi - 5) + (6 - 2\pi) - (12 - 4\pi)$$

$$\Rightarrow 4\pi - 11$$

3. **Ans (C)**

$$\Delta = \begin{vmatrix} -1 & 1 & 2 \\ 3 & -a & 5 \\ 2 & -2 & -a \end{vmatrix}$$

$$= -1(a^2 + 10) - 1(-3a - 10) + 2(-6 + 2a)$$

$$= -a^2 - 10 + 3a + 10 - 12 + 4a$$

$$\Delta = -a^2 + 7a - 12$$

$$\Delta = -[a^2 - 7a + 12]$$

$$\Delta = -[(a-3)(a-4)]$$

$$\Delta_1 = \begin{vmatrix} 0 & 1 & 2 \\ 1 & -a & 5 \\ 7 & -2 & -a \end{vmatrix}$$

$$= 0 - 1(-a - 35) + 2(-2 + 7a)$$

$$\Rightarrow a + 35 - 4 + 14a$$

$$15a + 31$$

Now $\Delta_1 = 15a + 31$

For inconsistent $\Delta = 0 \therefore a = 3, a = 4$

and for $a = 3$ and 4 $\Delta_1 \neq 0$

$n(S_1) = 2$

For infinite solution : $\Delta = 0$

and $\Delta_1 = \Delta_2 = \Delta_3 = 0$

Not possible

$\therefore n(S_2) = 0$

4. **Ans (B)**

$$P_1 : x - 2y - 2z + 1 = 0$$

$$P_2 : 2x - 3y - 6z + 1 = 0$$

$$\left| \frac{x - 2y - 2z + 1}{\sqrt{1 + 4 + 4}} \right| = \left| \frac{2x - 3y - 6z + 1}{\sqrt{2^2 + 3^2 + 6^2}} \right|$$

$$\frac{x - 2y - 2z + 1}{3} = \pm \frac{2x - 3y - 6z + 1}{7}$$

$$\text{Since } a_1a_2 + b_1b_2 + c_1c_2 = 20 > 0$$

$\therefore$ Negative sign will give

acute bisector

$$7x - 14y - 14z + 7 = -[6x - 9y - 18z + 3]$$

$$\Rightarrow 13x - 23y - 32z + 10 = 0$$

$$\left(-2, 0, -\frac{1}{2}\right) \text{ satisfy it}$$

5. **Ans (D)**

$$p \quad q \quad \sim p \quad \sim q \quad p \rightarrow q \quad \sim(p \rightarrow q) \quad q \rightarrow p \quad \sim(q \rightarrow p)$$

T	T	F	F	T	F	T	F
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	F	T	F

$$p \wedge \sim q \quad \sim p \rightarrow \sim q \quad p \rightarrow \sim q \quad \sim(p \rightarrow \sim q)$$

F	T	F	T
T	T	T	F
F	F	T	F
F	T	T	F

$$p \wedge \sim q \equiv \sim(p \rightarrow q)$$

6. **Ans (B)**

$$\text{Total ways of choosing square} = {}^{64}C_2$$

$$= \frac{64 \times 63}{2 \times 1} = 32 \times 63$$

ways of choosing two squares having common side

$$= 2(7 \times 8) = 112$$

$$\text{Required probability} = \frac{112}{32 \times 63} = \frac{16}{32 \times 9} = \frac{1}{18}$$

7. Ans (D)

$$x^2 dy + \left(y - \frac{1}{x}\right) dx = 0 : x > 0, y(1) = 1$$

$$x^2 dy + \frac{(xy - 1)}{x} dx = 0$$

$$x^2 dy = \frac{(xy - 1)}{x} dx$$

$$\frac{dy}{dx} = \frac{1 - xy}{x^3}$$

$$\frac{dy}{dx} = \frac{1}{x^3} - \frac{y}{x^2}$$

$$\frac{dy}{dx} = \frac{1}{x^2} \cdot y = \frac{1}{x^3}$$

$$\text{If } e^{\int \frac{1}{x^2} dx} = e^{-\frac{1}{x}}$$

$$ye^{-\frac{1}{x}} = \int \frac{1}{x^3} \cdot e^{-\frac{1}{x}}$$

$$ye^{-\frac{1}{x}} = e^{-x} \left(1 + \frac{1}{x}\right) + C$$

$$1 \cdot e^{-1} = e^{-1}(2) + C$$

$$C = -e^{-1} = -\frac{1}{e}$$

$$ye^{-\frac{1}{x}} = e^{-\frac{1}{x}} \left(1 + \frac{1}{x}\right) - \frac{1}{e}$$

$$y \left(\frac{1}{2}\right) = 3 - \frac{1}{e} \times e^2$$

$$y \left(\frac{1}{2}\right) = 3 - e$$

8. Ans (A)

$$2 \cos x \left(4 \sin \left(\frac{\pi}{4} + x\right) \sin \left(\frac{\pi}{4} - x\right) - 1\right) = 1$$

$$2 \cos x \left(4 \left(\sin^2 \frac{\pi}{4} - \sin^2 x\right) - 1\right) = 1$$

$$2 \cos x \left(4 \left(\frac{1}{2} - \sin^2 x\right) - 1\right) = 1$$

$$2 \cos x (2 - 4 \sin^2 x - 1) = 1$$

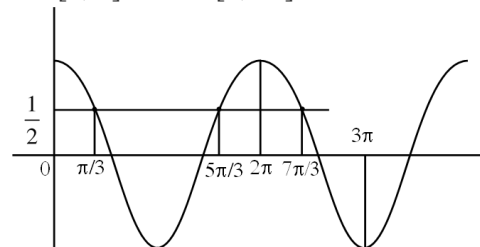
$$2 \cos x (1 - 4 \sin^2 x) = 1$$

$$2 \cos x (4 \cos^2 x - 3) = 1$$

$$4 \cos^3 x - 3 \cos x = \frac{1}{2}$$

$$\cos 3x = \frac{1}{2}$$

$$x \in [0, \pi] \therefore 3x \in [0, 3\pi]$$



9. Ans (A)

$$f(x) = x^3 - 6x^2 + ax + b$$

$$f(2) = 8 - 24 + 2a + b = 0$$

$$2a + b = 16 \dots (1)$$

$$f(4) = 64 - 96 + 4a + b = 0$$

$$4a + b = 32 \dots (2)$$

Solving (1) and (2)

$$a = 8, b = 0$$

$$f(x) = x^3 - 6x^2 + 8x$$

$$f(x) = 3x^2 - 12x + 8$$

$$f'(x) = 6x - 12$$

$\Rightarrow f'(x)$ is $\uparrow$ for $x > 2$, and $f'(x)$ is $\downarrow$ for $x < 2$

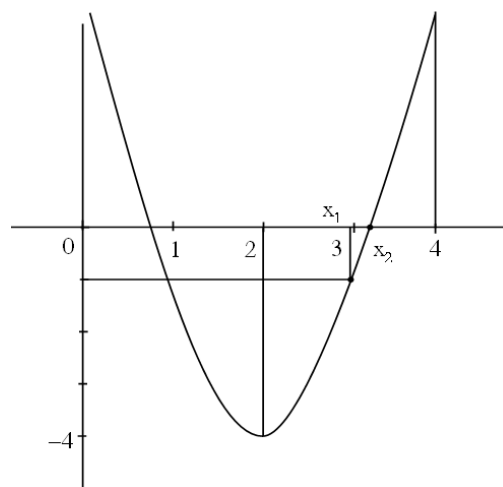
$$f(2) = 12 - 24 + 8 = -4$$

$$f(4) = 48 - 48 + 8 = 8$$

$$f(x) = 3x^2 - 12x + 8$$

vertex $(2, -4)$

$$f(2) = -4, f(4) = 8, f(3) = 27 - 36 + 8$$



$$f(x_1) = -1, \text{ then } x_1 = 3$$

$$f(x_2) = 0$$

Again $f'(x) < 0$ for $x \in (2, x_4)$

$$f'(x) > 0 \text{ for } x \in (x_4, 4)$$

$$x_4 \in (3, 4)$$

$$f(x) = x^3 - 6x^2 + 8x$$

$$f(3) = 27 - 54 + 24 = -3$$

$$f(4) = 64 - 96 + 32 = 0$$

For $x_4(3, 4)$

$$f(x_4) < -3\sqrt{3}$$

$$\text{and } f'(x_3) > -4$$

$$2f'(x_3) > -8$$

$$\text{So, } 2f'(x_3) = \sqrt{3} f(x_4)$$

10. Ans (C)

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$J_{6+i,3} - J_{i+3,3}; i \leq j$$

$$\Rightarrow \int_0^{\frac{1}{2}} \frac{x^{6+i}}{x^3-1} - \int_0^{\frac{1}{2}} \frac{x^{i+3}}{x^3-1}$$

$$\int_0^{1/2} \frac{x^{i+3}(x^3-1)}{x^3-1}$$

$$\frac{x^{3+i+1}}{3+i+1} = \left(\frac{x^{4+i}}{4+i} \right)_0^{1/2}$$

$$a_{ij} = J_{6+i,3} - J_{i+3,3} = \frac{\left(\frac{1}{2}\right)^{4+i}}{4+i}$$

$$a_{11} = \frac{\left(\frac{1}{2}\right)^5}{5} = \frac{1}{5 \cdot 2^5}$$

$$a_{12} = \frac{1}{5 \cdot 2^5}$$

$$a_{13} = \frac{1}{5 \cdot 2^5}$$

$$a_{22} = \frac{1}{6 \cdot 2^6}$$

$$a_{23} = \frac{1}{6 \cdot 2^6}$$

$$a_{33} = \frac{1}{7 \cdot 2^7}$$

$$A = \begin{bmatrix} \frac{1}{5 \cdot 2^5} & \frac{1}{5 \cdot 2^5} & \frac{1}{5 \cdot 2^5} \\ 0 & \frac{1}{6 \cdot 2^6} & \frac{1}{6 \cdot 2^6} \\ 0 & 0 & \frac{1}{7 \cdot 2^7} \end{bmatrix}$$

$$|A| = \frac{1}{5 \cdot 2^5} \left[\frac{1}{6 \cdot 2^6} \times \frac{1}{7 \cdot 2^7} \right]$$

$$|A| = \frac{1}{210 \cdot 2^{18}}$$

$$|\text{adj} A^{-1}| = |A^{-1}|^{n-1} = |A^{-1}|^2 = \frac{1}{(|A|)^2}$$

$$\Rightarrow (210 \cdot 2^{18})^2$$

$$(105)^2 \times 2^{38}$$

11. Ans (A)

$$A = \int_0^{\pi/2} ((\sin x + \cos x) - |\cos x - \sin x|) dx$$

$$A = \int_0^{\pi/2} ((\sin x + \cos x) - (\cos x - \sin x)) dx$$

$$+ \int_{\pi/4}^{\pi/2} ((\sin x + \cos x) - (\sin x - \cos x)) dx$$

$$A = 2 \int_0^{\pi/2} \sin x dx + 2 \int_{\pi/4}^{\pi/2} \cos x dx$$

$$A = -2 \left(\frac{1}{\sqrt{2}} - 1 \right) + 2 \left(1 - \frac{1}{\sqrt{2}} \right)$$

$$A = 4 - 2\sqrt{2} = 2\sqrt{2}(\sqrt{2} - 1)$$

12. Ans (C)

$$3y - 2z - 1 = 0 = 3x - z + 4$$

$$3y - 2z - 1 = 0 \quad \text{D.R's} \Rightarrow (0, 3, -2)$$

$$3x - z + 4 = 0 \quad \text{D.R's} \Rightarrow (3, -1, 0)$$

Let DR's of given line are a, b, c

$$\text{Now } 3b - 2c = 0 \text{ \& } 3a - c = 0$$

$$\therefore 6a = 3b = 2c$$

$$a : b : c = 3 : 6 : 9$$

Any pt on line

$$3K - 1, 6K + 1, 9K + 1$$

$$\text{Now } 3(3K - 1) + 6(6K + 1) + 9(9K + 1) = 0$$

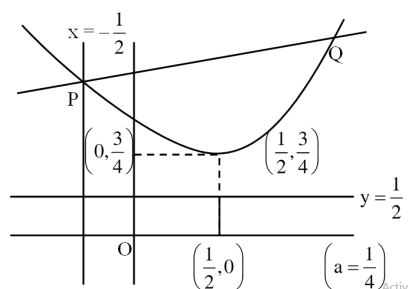
$$\Rightarrow K = \frac{1}{3}$$

$$\text{Point on line} \Rightarrow (0, 3, 4)$$

$$\text{Given point } (2, -1, 6)$$

$$\Rightarrow \text{Distance} = \sqrt{4 + 16 + 4} = 2\sqrt{6}$$

13. Ans (B)



$$\left(y - \frac{3}{4}\right) = \left(x - \frac{1}{2}\right)^2 \quad \dots (1)$$

For $x = -\frac{1}{2}$

$$y - \frac{3}{4} = 1 \Rightarrow y = \frac{7}{4} \Rightarrow P\left(-\frac{1}{2}, \frac{7}{4}\right)$$

Now $y' = 2\left(x - \frac{1}{2}\right)$ At $x = -\frac{1}{2}$

$$\Rightarrow m_T = -2, m_N = \frac{1}{2}$$

Equation of Normal is

$$y - \frac{7}{4} = \frac{1}{2}\left(x + \frac{1}{2}\right)$$

$$y = \frac{x}{2} + 2$$

Now put y in equation (1)

$$\frac{x}{2} + 2 - \frac{3}{4} = \left(x - \frac{1}{2}\right)^2$$

$$\Rightarrow x = 2 \text{ \& } -\frac{1}{2}$$

$$\Rightarrow Q(2, 3)$$

$$\text{Now } (PQ)^2 = \frac{125}{16}$$

14. Ans (A)

Consider the equation $x^2 + ax + b = 0$

If has two roots (not necessarily real α & β)

Either $\alpha = \beta$ or $\alpha \neq \beta$

Case (1) If $\alpha = \beta$, then it is repeated root.

Given that $\alpha^2 - 2$ is also a root

$$\text{So, } \alpha = \alpha^2 - 2 \Rightarrow (\alpha + 1)(\alpha - 2) = 0$$

$$\Rightarrow \alpha = -1 \text{ or } \alpha = 2$$

$$\text{When } \alpha = -1 \text{ then } (a, b) = (2, 1)$$

$$\alpha = 2 \text{ then } (a, b) = (-4, 4)$$

Case (2) If $\alpha \neq \beta$ Then

$$\text{(I) } \alpha = \alpha^2 - 2 \text{ and } \beta = \beta^2 - 2$$

$$\text{Here } (\alpha, \beta) = (2, -1) \text{ or } (-1, 2)$$

$$\text{Hence } (a, b) = -(\alpha + \beta), \alpha\beta$$

$$= (-1, -2)$$

$$\text{(II) } \alpha = \beta^2 - 2 \text{ and } \beta = \alpha^2 - 2$$

$$\text{Then } \alpha - \beta = \beta^2 - \alpha^2 = (\beta - \alpha)(\beta + \alpha)$$

$$\text{Since } \alpha \neq \beta \text{ we get } \alpha + \beta = \beta^2 + \alpha^2 - 4$$

$$\alpha + \beta = (\alpha + \beta)^2 - 2\alpha\beta - 4$$

$$\text{Thus } -1 = 1 - 2\alpha\beta - 4 \text{ which implies}$$

$$\alpha\beta = -1 \text{ Therefore } (a, b) = -(\alpha + \beta), \alpha\beta$$

$$= (1, -1)$$

$$\text{(III) } \alpha = \alpha^2 - 2 = \beta^2 - 2 \text{ and } \alpha \neq \beta$$

$$\Rightarrow \alpha = -\beta$$

$$\text{Thus } \alpha = 2, \beta = -2$$

$$\alpha = -1, \beta = 1$$

$$\text{Therefore } (a, b) = (0, -4) \text{ \& } (0, -1)$$

$$\text{(IV) } b = \alpha^2 - 2 = \beta^2 - 2 \text{ and } \alpha \neq \beta \text{ is same as (III)}$$

Therefore we get 6 pairs of (a, b)

Which are $(2, 1), (-4, 4), (-1, -2), (1, -1), (0, -4)$

15. **Ans (A)**

$$\text{Let } T_r = r(n-r)$$

$$T_r = nr - r^2$$

$$\Rightarrow S_n = \sum_{r=1}^n T_r = \sum_{r=1}^n (nr - r^2)$$

$$S_n = \frac{n \cdot (n) \cdot (n+1)}{2} - \frac{n(n+1)(2n+1)}{6}$$

$$S_n = \frac{n(n-1)(n+1)}{6}$$

$$\begin{aligned} \text{Now } \sum_{r=4}^{\infty} \left(\frac{2S_n}{n!} - \frac{1}{(n-2)!} \right) \\ = \sum_{r=4}^{\infty} \left(2 \cdot \frac{n(n-1)(n+1)}{6 \cdot n(n-1)(n-2)!} - \frac{1}{(n-2)!} \right) \\ = \sum_{r=4}^{\infty} \left(\frac{1}{3} \left(\frac{n-2+3}{(n-2)!} \right) - \frac{1}{(n-2)!} \right) \\ = \sum_{r=4}^{\infty} \frac{1}{3} \cdot \frac{1}{(n-3)!} = \frac{1}{3}(e-1) \end{aligned}$$

16. **Ans (C)**

$$\text{Total Number of Triangles} = {}^{15}C_3$$

$$i+j+k=15 \text{ (Given)}$$

5 Cases

i	j	k	4 Cases	3 Cases	1 Cases
1	2	12	i	j	k
1	3	11	2	3	10
1	4	10	2	4	9
1	5	9	2	5	8
1	6	8	2	6	7
			3	4	8
			3	5	7
			4	5	6

Number of Possible triangles using the vertices P_i, P_j, P_k such that $i+j+k \neq 15$ is equal to ${}^{15}C_3 - 12 = 443$

17. **Ans (D)**

$$f(x) = \log_{\sqrt{5}}$$

$$\left(3 + \cos\left(\frac{3\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} + x\right) + \cos\left(\frac{\pi}{4} - x\right) - \cos\left(\frac{3\pi}{4} - x\right) \right)$$

$$f(x) = \log_{\sqrt{5}} \left[3 + 2 \cos\left(\frac{\pi}{4}\right) \cos(x) - 2 \sin\left(\frac{3\pi}{4}\right) \sin(x) \right]$$

$$f(x) = \log_{\sqrt{5}} [3 + \sqrt{2}(\cos x - \sin x)]$$

$$\text{Since } -\sqrt{2} \leq \cos x - \sin x \leq \sqrt{2}$$

$$\Rightarrow \log_{\sqrt{5}} [3 + \sqrt{2}(-\sqrt{2})] \leq f(x) \leq \log_{\sqrt{5}} [3 + \sqrt{2}(\sqrt{2})]$$

$$\Rightarrow \log_{\sqrt{5}} (1) \leq f(x) \leq \log_{\sqrt{5}} (5)$$

So Range of $f(x)$ is $[0, 2]$

18. **Ans (B)**

$$\sum_{n=1}^{20} \frac{1}{a_n a_{n+1}} = \sum_{n=1}^{20} \frac{1}{a_n (a_n + d)}$$

$$= \frac{1}{d} \sum_{n=1}^{20} \left(\frac{1}{a_n} - \frac{1}{a_n + d} \right)$$

$$\Rightarrow \frac{1}{d} \left(\frac{1}{a_1} - \frac{1}{a_{21}} \right) = \frac{4}{9} \text{ (Given)}$$

$$\Rightarrow \frac{1}{d} \left(\frac{a_{21} - a_1}{a_1 a_{21}} \right) = \frac{4}{9}$$

$$\Rightarrow \frac{1}{d} \left(\frac{a_1 + 20d - a_1}{a_1 a_2} \right) = \frac{4}{9} \Rightarrow a_1 a_2 = 45$$

.... (1)

$$\text{Now sum of first 21 terms} = \frac{21}{2} (2a_1 + 20d) = 189$$

$$\Rightarrow a_1 + 10d = 9 \text{ ... (2)}$$

For equation (1) & (2) we get

$$a_1 = 3 \text{ \& } d = \frac{3}{5}$$

OR

$$a_1 = 15 \text{ \& } d = -\frac{3}{5}$$

$$\text{So, } a_6 a_{16} = (a_1 + 5d)(a_1 + 15d)$$

$$\Rightarrow a_6 a_{16} = 72$$

19. **Ans (D)**

$$f(x) = x + \int_0^{\frac{\pi}{2}} \sin x \cos y f(y) dy$$

$$f(x) = x + \sin x \underbrace{\int_0^{\pi/2} \cos y f(y) dy}_K$$

$$\Rightarrow f(x) = x + K \sin x$$

$$\Rightarrow f(y) = y + K \sin y$$

$$\text{Now } K = \int_0^{\frac{\pi}{2}} \cos y (y + K \sin y) dy$$

$$K = \int_0^{\frac{\pi}{2}} y \cos y dy + \int_0^{\frac{\pi}{2}} \cos y \sin y dy$$

Apply IBP Put $\sin y = t$

$$K = (y \sin y)_0^{\pi/2} - \int_0^{\pi/2} \sin y dy + K \int_0^1 t dt$$

$$\Rightarrow K = \frac{\pi}{2} - 1 + K \left(\frac{1}{2} \right)$$

$$\Rightarrow K = \pi - 2$$

$$\text{So } f(x) = x + (\pi - 2) \sin x$$

20. Ans (B)

The point of intersection of the curves $\frac{x^2}{9} + \frac{y^2}{1} = 1$ and $x^2 + y^2 = 3$ in the first quadrant is $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right)$

quadrant is $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right)$

Now slope of tangent to the ellipse $\frac{x^2}{9} + \frac{y^2}{1} = 1$ at $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right)$ is $m_1 = -\frac{1}{3\sqrt{3}}$

And slope of tangent to the circle at $\left(\frac{3}{2}, \frac{\sqrt{3}}{2}\right)$

is $m_2 = -\sqrt{3}$

So, if angle between both curves is θ then

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{-\frac{1}{3\sqrt{3}} + \sqrt{3}}{1 + \left(-\frac{1}{3\sqrt{3}}\right)(-\sqrt{3})} \right|$$

$$= \frac{2}{\sqrt{3}}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (781)

x	-2	-1	3	4	6
P(X = x)	$\frac{1}{5}$	a	$\frac{1}{3}$	$\frac{1}{5}$	b

$$\bar{X} = 2.3$$

$$-a + 6b = \frac{9}{10} \quad \dots(1)$$

$$\sum P_i = \frac{1}{5} + a + \frac{1}{3} + \frac{1}{5} + b = 1$$

$$a + b = \frac{4}{15} \quad \dots(2)$$

From equation (1) and (2)

$$a = \frac{1}{10}, b = \frac{1}{6}$$

$$\sigma^2 = \sum p_i x_i^2 - (\bar{X})^2$$

$$\frac{1}{5}(4) + a(1) + \frac{1}{3}(9)$$

$$+ \frac{1}{5}(16) + b(36) - (2.3)^2$$

$$= \frac{4}{5} + a + 3 + \frac{16}{5} + 36b - (2.3)^2$$

$$= 4 + a + 3 + 36b - (2.3)^2$$

$$= 7 + a + 36b - (2.3)^2$$

$$= 7 + \frac{1}{10} + 6 - (2.3)^2$$

$$= 13 + \frac{1}{10} - \left(\frac{23}{10}\right)^2$$

$$= \frac{131}{10} - \left(\frac{23}{10}\right)^2$$

$$= \frac{1310 - (23)^2}{100}$$

$$= \frac{1310 - 529}{100}$$

$$\sigma^2 = \frac{781}{100}$$

$$100\sigma^2 = 781$$

2. Ans (7)

$$f(n) = x^6 + 2x^4 + x^3 + 2x + 3$$

$$\lim_{x \rightarrow 1} \frac{x^n f(1) - f(x)}{x - 1} = 44$$

$$\lim_{x \rightarrow 1} \frac{9x^n - (x^6 + 2x^4 + x^3 + 2x + 3)}{x - 1} = 44$$

$$\lim_{x \rightarrow 1} \frac{9nx^{n-1} - (6x^5 + 8x^3 + 3x^2 + 2)}{1} = 44$$

$$\Rightarrow 9n - (19) = 44$$

$$\Rightarrow 9n = 63$$

$$\Rightarrow n = 7$$

3. Ans (5)

$$|z - 2 - 2i| \leq 1$$

$$|x + iy - 2 - 2i| \leq 1$$

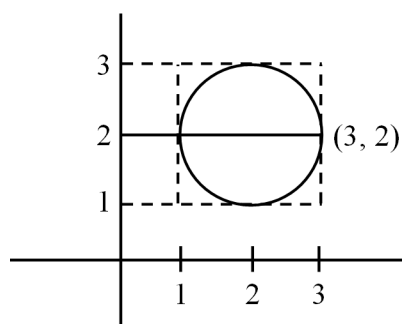
$$|(x - 2) + i(y - 2)| \leq 1$$

$$(x - 2)^2 + (y - 2)^2 \leq 1$$

$$|3iz + 6|_{\max} \text{ at } a + ib$$

$$|3i| \left| z + \frac{6}{3i} \right|$$

$$3|z - 2i|_{\max}$$

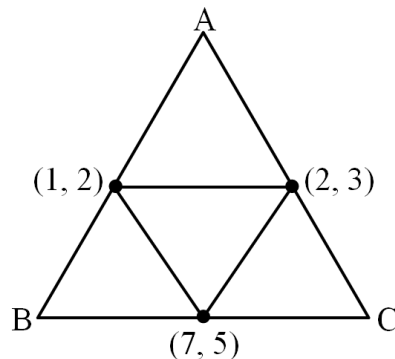


From Figure maximum distance at $3 + 2i$

$$a + ib = 3 + 2i = a + b = 3 + 2 = 5$$

4. Ans (6)

intersection point of give lines are $(1, 2)$, $(7, 5)$, $(2, 3)$



$$\Delta = \frac{1}{2} \begin{vmatrix} 1 & 2 & 1 \\ 7 & 5 & 1 \\ 2 & 3 & 1 \end{vmatrix}$$

$$= \frac{1}{2} [1(5 - 3) - 2(7 - 2) + 1(21 - 10)]$$

$$= \frac{1}{2} [2 - 10 + 11]$$

$$\Delta DEF = \frac{1}{2} (3) = \frac{3}{2}$$

$$\Delta ABC = 4 \Delta DEF = 4 \left(\frac{3}{2} \right) = 6$$

5. Ans (26)

$$kf(k) + 2 = \lambda (x - 2)(x - 3)(x - 4)(x - 5) \dots (1)$$

put $x = 0$

$$\text{we get } \lambda = \frac{1}{60}$$

Now put λ in equation (1)

$$\Rightarrow kf(k) + 2 = \frac{1}{60} (x - 2)(x - 3)(x - 4)(x - 5)$$

Put $x = 10$

$$\Rightarrow 10f(10) + 2 = \frac{1}{60} (8)(7)(6)(5)$$

$$\Rightarrow 52 - 10f(10) = 52 - 26 = 26$$

6. Ans (77)

FARMER (6)

A, E, F, M, R, R

A					
E					
F	A	E			
F	A	M			
F	A	R	E		
F	A	R	M	E	R

$$\frac{5}{2} - 4 = 60 - 24 = 36$$

$$\frac{3}{2} - 2 = 3 - 2 = 1$$

$$= 1$$

$$= 2$$

$$= 1$$

$$77$$

7. Ans (924)

$$(x + y)^n \Rightarrow 2^n = 4096 \quad 2^{10} = 1024 \times 2$$

$$\Rightarrow 2^n = 2^{12} \quad 2^{11} = 2048$$

$$n = 12 \quad 2^{12} = 4096$$

$${}^{12}C_6 = \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$$= 11 \times 3 \times 4 \times 7$$

$$= 924$$

8. Ans (1494)

$$\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$$

$$\vec{c} = 3\hat{i} + 2\hat{j} - \hat{k}$$

$$\vec{v} = x\vec{a} + y\vec{b} \quad \vec{v} \cdot (3\hat{i} + 2\hat{j} - \hat{k}) = 0$$

$$\vec{v} \cdot \hat{a} = 19$$

$$\vec{v} = \lambda \vec{c} \times (\vec{a} \times \vec{b})$$

$$\vec{v} = \lambda \left[(\vec{c} \cdot \vec{b}) \vec{a} - (\vec{c} \cdot \vec{a}) \vec{b} \right]$$

$$= \lambda [(3+4+1)(2\hat{i} - \hat{j} + 2\hat{k}) - \left(\frac{6-2-2}{2} \right)$$

$$(\hat{i} + 2\hat{j} + \hat{k})$$

$$= \lambda [16\hat{i} - 8\hat{j} + 16\hat{k} - 2\hat{i} - 4\hat{j} + 2\hat{k}]$$

$$\vec{v} = \lambda [14\hat{i} - 12\hat{j} + 18\hat{k}]$$

$$\lambda [14\hat{i} - 12\hat{j} + 18\hat{k}] \cdot \frac{(2\hat{i} - \hat{j} + 2\hat{k})}{\sqrt{4+1+4}} = 19$$

$$\lambda \frac{[28 + 12 + 36]}{3} = 19$$

$$\lambda \left(\frac{76}{3} \right) = 19$$

$$4\lambda = 3 \Rightarrow \lambda = \frac{3}{4}$$

$$|2\vec{v}| = \left| 2 \times \frac{3}{4} (14\hat{i} - 12\hat{j} + 18\hat{k}) \right|^2$$

$$\frac{9}{4} \times 4 (7\hat{i} - 6\hat{j} + 9\hat{k})^2$$

$$= 9 (49 + 36 + 81)$$

$$= 9 (166)$$

$$= 1494$$

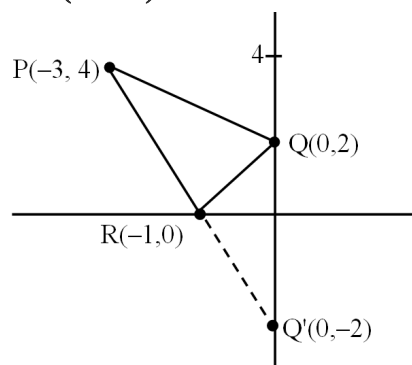
9. Ans (2)

$$f(x) = [x] |x^2 - 1| + \sin \frac{\pi}{[x+3]} - [x+1]$$

$$f(x) = \begin{cases} 3 - 2x^2, & -2 < x < -1 \\ x^2, & -1 \leq x < 0 \\ \frac{\sqrt{3}}{2} + 1, & 0 \leq x < 1 \\ x^2 + 1 + \frac{1}{\sqrt{2}}, & 1 \leq x < 2 \end{cases}$$

discontinuous at $x = 0, 1$

10. Ans (1250)



$$50(PR^2 + RQ^2)$$

$$= 50 (20 + 5)$$

$$= 50(25)$$

$$= 1250$$